Predicting Students' Science Average Based on Study Habits and Personal Characteristics

Research Question: Can study habits and personal characteristics predict students' Science Average?

Hypothesis: H_0 (Null Hypothesis): Students aspiring to STEM careers have the same average Science Average as students aspiring to Non-STEM careers.

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Dataset Selection

- To answer the research question, the **Score 6K Students** dataset was selected. The dataset originally contained approximately **6,000 student records** and included demographic information, study habits, academic performance and career aspirations.
- **Dependent Variable (Y):** Science Average
- **Independent Variables (X):**Weekly Self-Study Hours , Absence Days , Gender , Part-Time Job, Extracurricular Activities
- The selected variables were used to investigate whether personal characteristics and study habits can predict students' Science Average.

Data Cleaning and Preparation

- Before performing any statistical analysis, the dataset was cleaned and prepared. The following preprocessing steps were performed:
- Removed duplicate observations;
- Removed variables that were not relevant to the research question;
- Created the **Science Average** variable by calculating the average of **Biology, Chemistry, Math and Physics** scores;
- Created the **Career Category** variable by grouping careers into **STEM** and **Non-STEM;**
- Created the STEM binary variable (0 = Non-STEM, 1 = STEM);
- Converted categorical variables into numeric values to make them suitable for regression analysis: Gender, Part-Time Job, Extracurricular Activities;
- These preprocessing steps ensured that the dataset was suitable for descriptive statistics, hypothesis testing and multiple linear regression.

Descriptive Statistics

- Before performing any statistical tests, descriptive statistics were calculated to better understand the distribution of the main continuous variables.
- The following variables were analyzed: Science Average, Weekly Self-Study Hours, Absence Days.
- For each variable, the following statistics were calculated: Mean, Median, Standard Deviation, Min, Max, Q1, Q3, IQR, Lower Fence, Upper Fence
- These statistics helped identify the center, spread and potential outliers within the dataset before proceeding with further analysis.

	A	B	C	D	E	F	G	H	I	J	K	L	M
	absence_days					weekly_self_study_hours					science_average		
	Mean	3.666166917				Mean	17.76011994				Mean	81.09133	
	Median	3				Median	18				Median	81.25	
	St_dev	2.628782941				St_dev	12.12833166				St_dev	7.591661	
	Q1	2				Q1	5				Q1	76.25	
	Q3	5				Q3	28				Q3	86.5	
	IQR	3				IQR	23				IQR	10.25	
	Lower_fence	-2.5				Lower_fence	-29.5				Lower_fence	60.875	
	Upper_fence	9.5				Upper_fence	62.5				Upper_fence	101.875	
	Min	0				Min	0				Min	51.5	
	Max	10				Max	50				Max	97.5	

Categorical Summary

- After analyzing the continuous variables, the categorical variables were summarized using Pivot Tables. The following variables were analyzed: Gender, Part-Time Job, Extracurricular Activities, Career Category (STEM / Non-STEM)
- For each category, the following information was calculated: Number of students, percentage of students, mean science average, mean weekly self-study hours and standard deviation.
- These summaries provided an overview of the dataset and allowed comparisons between different student groups before building the prediction model.

Gender	Count	Percentage	Average of Science Average	Average of Study Hours
female	1002	50.10%	80.77	16.82
male	998	49.90%	81.42	18.69
Total	**2000**	**100.00%**	**81.09**	**17.76**

Part-time Job	Count	Percentage	Average of Science Average	Average of Study Hours
FALSE	1684	84.20%	81.65	18.88
TRUE	316	15.80%	78.09	11.77
Total	**2000**	**100.00%**	**81.09**	**17.76**

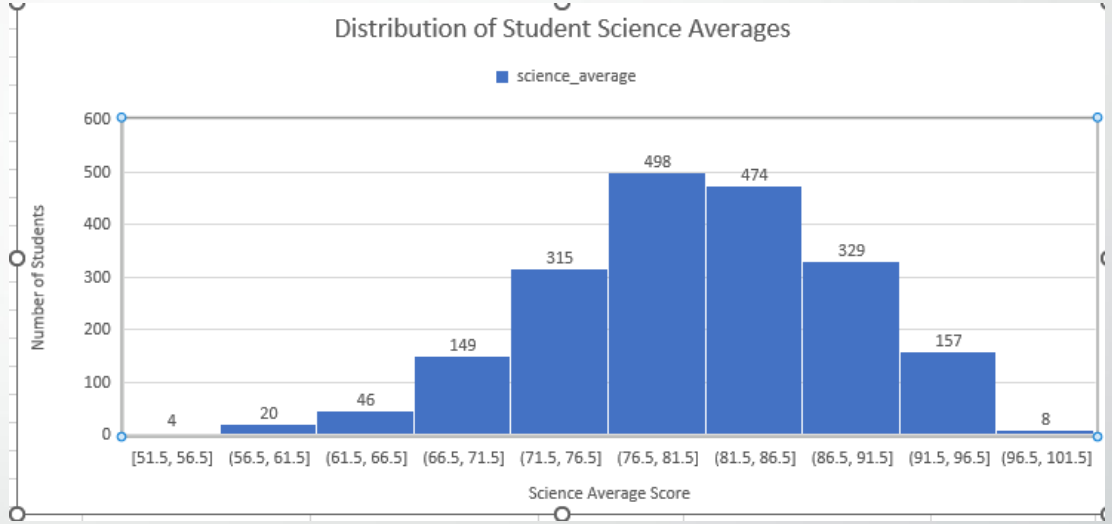
Extracurriculu	Count	Percentage	Average of Science Average	Average of Study Hours
FALSE	1592	79.60%	81.22	18.08
TRUE	408	20.40%	80.60	16.50
Total	**2000**	**100.00%**	**81.09**	**17.76**

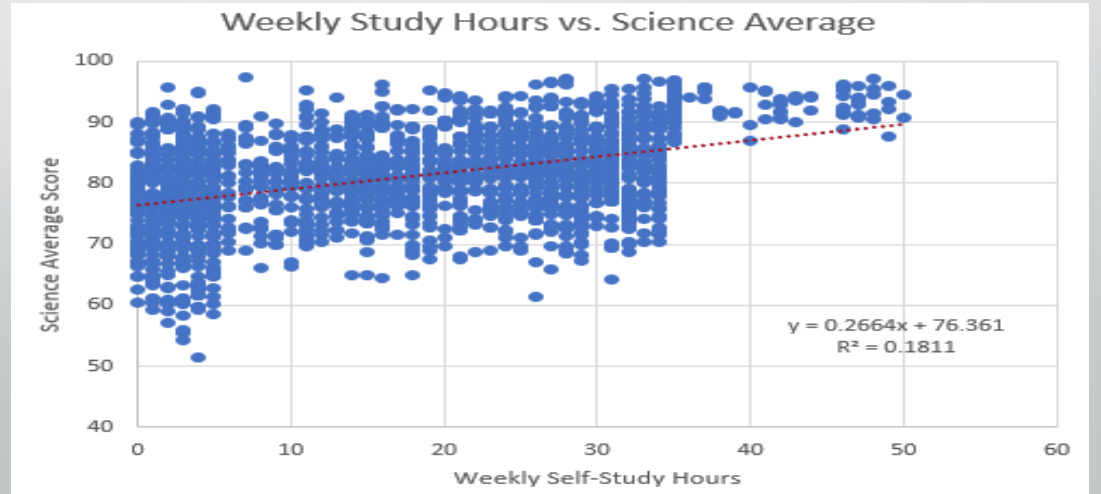
Career Aspire	Count	Percentage	Average of Science Average	Average of Study Hours	StDev of science_average	StDev of self_study
Non-STEM	1396	69.80%	79.05	14.74	7.31886841	11.32118869
STEM	604	30.20%	85.81	24.72	5.939271762	11.03113247
Total	**2000**	**100.00%**	**81.09**	**17.76**	**7.593558422**	**12.1296036**

Exploratory Data Analysis (EDA)

- Exploratory Data Analysis (EDA) was performed to investigate the relationships between the independent variables and Science Average before building the prediction model.
- The following visualizations were created: Distribution of Science Average, Weekly Self-Study Hours vs Science Average, Mean Science Average: STEM vs. Non-STEM Aspirations, Absence Days vs Science Average.

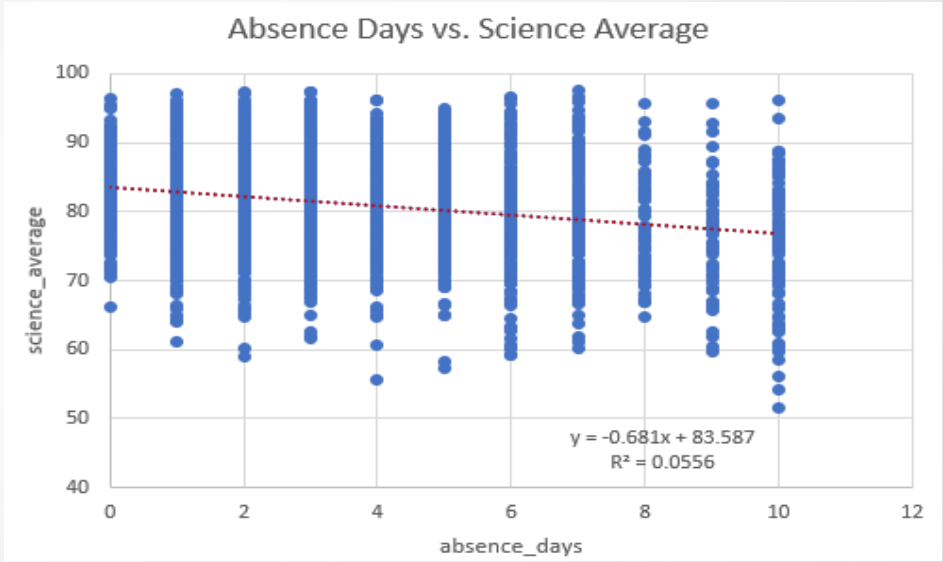
- The histogram showed that most students' Science Average values were concentrated around the center, while fewer students had very low or very high scores.
- The scatter plot of Weekly Self-Study Hours showed a positive relationship, indicating that students who study more tend to achieve higher Science Average values.

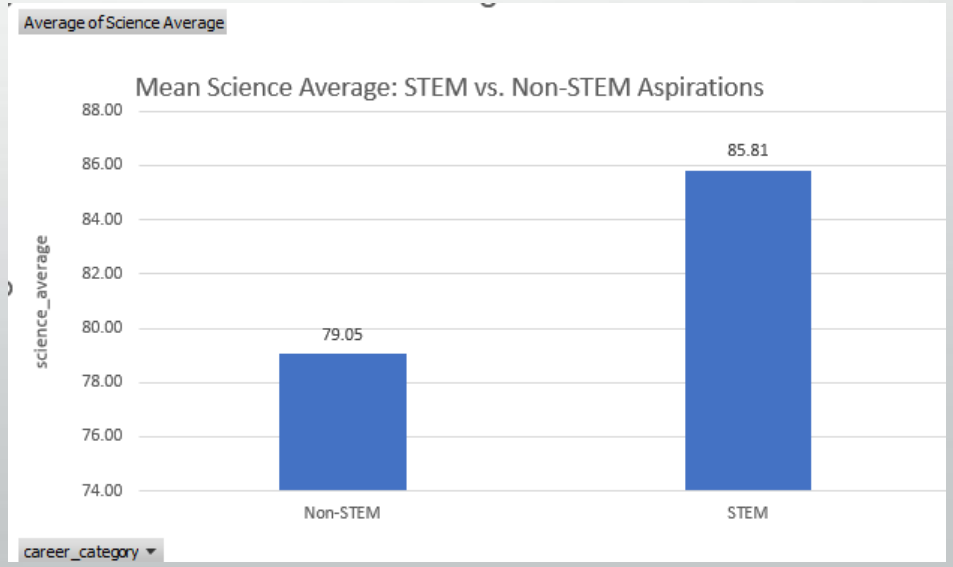




Exploratory Data Analysis (II)

- The scatter plot of Absence Days showed a negative relationship, indicating that students with more absences tend to have lower Science Average values.
- Trendlines and R^2 values were included to evaluate the strength of these relationships.
- Additional visualizations were created to further understand the relationships within the dataset.
- The comparison between STEM and Non-STEM students showed that students aspiring to STEM careers achieved a higher mean Science Average than students aspiring to Non-STEM careers.





Exploratory Data Analysis (III)

- A correlation matrix was also created to examine the relationships among all variables included in the regression model.
- The correlation matrix showed that Weekly Self-Study Hours had a positive correlation with Science Average, while Absence Days showed a negative correlation. The remaining variables demonstrated weaker relationships.

	absence_days	*weekly_self_study_hours*	*gender_numeric*	*part_time_numeric*	*extracurricular_numeric*	*science_average*
absence_days	1					
weekly_self_study_hour	-0.286085623	1				
gender_numeric	-0.029357359	0.077104474	1			
part_time_numeric	0.206360656	-0.213962621	-0.023808723	1		
extracurricular_numeric	0.029018652	-0.052515108	-0.00643227	-0.008382132	1	
science_average	-0.235798016	0.42555502	0.04273547	-0.171332351	-0.03272584	1

These findings provided an initial understanding of the data before performing statistical hypothesis testing.

Hypothesis Testing

- To determine whether STEM students achieve higher Science Average scores than Non-STEM students, an independent samples Welch's t-test was performed.
- **Null Hypothesis (H_0):**
  Students aspiring to STEM careers have the same average Science Average as students aspiring to Non-STEM careers.
- **Alternative Hypothesis (H_1):**
  Students aspiring to STEM careers have a higher average Science Average than students aspiring to Non-STEM careers.
- Results: Mean Science Average: STEM = **85.81**, Non-STEM = **79.04**, p-value = **1.93E-90**
- Since the p-value is less than 0.05, the null hypothesis was rejected.

The results indicate that students aspiring to STEM careers have a significantly higher Science Average than students aspiring to Non-STEM careers.

Multiple Linear Regression

- Multiple Linear Regression was performed to answer the research question by evaluating whether personal characteristics and study habits can predict students' Science Average. The regression statistics showed:
- Multiple R = **0.447**, indicating a moderate positive relationship between the predicted and actual Science Average values.
- R^2 = **0.200**, meaning that approximately 20% of the differences in Science Average are explained by the independent variables included in the model.
- Adjusted R^2 = **0.198**, which is very close to R^2, indicating that the predictors contribute useful information without unnecessarily increasing the complexity of the model.
- Standard Error = **6.80**, indicating that the predicted Science Average differs from the observed value by approximately 6.8 points on average.

Overall, the regression statistics indicate that the model has a moderate predictive ability.

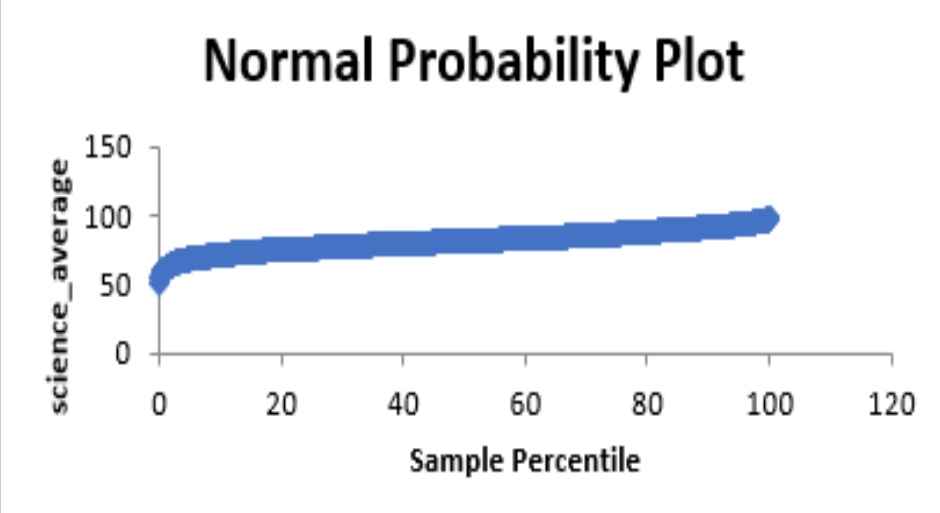
Multiple Linear Regression Results

- **Regression Results:** The ANOVA table was used to evaluate the regression model as a whole. The F-statistic was **99.45**, and the Significance F was **8.40×10^{-94}**, which is far below 0.05.
- This indicates that the regression model is statistically significant and that at least one independent variable contributes significantly to predicting Science Average.
- The coefficients table showed that Weekly Self-Study Hours, Absence Days and Part-Time Job are significant predictors of Science Average, while Gender and Extracurricular Activities are not statistically significant predictors in this model.

	Coefficients	*Standard Error*	*t Stat*	*P-value*
Intercept	78.2886536	0.432775921	180.8988205	0
absence_days	-0.32751263	0.061137443	-5.356989295	9.43953E-08
weekly_self_study_hours	0.236375487	0.013320543	17.74518462	1.5701E-65
gender_numeric	0.131074402	0.30512352	0.429578165	0.667548937
part_time_numeric	-1.395005576	0.432239472	-3.227390529	0.001269513
extracurricular_numeric	-0.19052087	0.378106761	-0.503881151	0.614400561

Model Diagnostics

- After building the regression model, diagnostic tests were performed to verify that the model satisfied the assumptions of multiple linear regression.
- The Normal Probability Plot was used to assess whether the distribution of Science Average was approximately normal. The plot showed that the observations followed a generally smooth pattern, with some deviations at the tails.
- Variance Inflation Factor (VIF) was calculated for each independent variable to assess multicollinearity. All VIF values were close to 1, indicating that the independent variables are not highly correlated with one another. Therefore, multicollinearity is not present, and the regression coefficients can be interpreted with confidence.



VIF	Value
VIF(extra_activities)	1.003500994
VIF(part_time_job)	1.074394337
VIF(gender)	1.006086514
VIF(self_study)	1.12788684
VIF (absencedays)	1.116384384

Conclusion

- The objective of this project was to determine whether personal characteristics and study habits can predict students' Science Average. The results showed that:
- Weekly Self-Study Hours are a significant positive predictor of Science Average;
- Absence Days are a significant negative predictor of Science Average;
- Having a Part-Time Job is a significant negative predictor of Science Average;
- Gender and Extracurricular Activities were not statistically significant predictors.
- The regression model explained approximately **20%** of the differences in students' Science Average, indicating that additional factors not included in the dataset also influence academic performance.

Overall, the results suggest that personal characteristics and study habits can predict students' Science Average, although other variables outside the model also contribute to students' academic achievement.